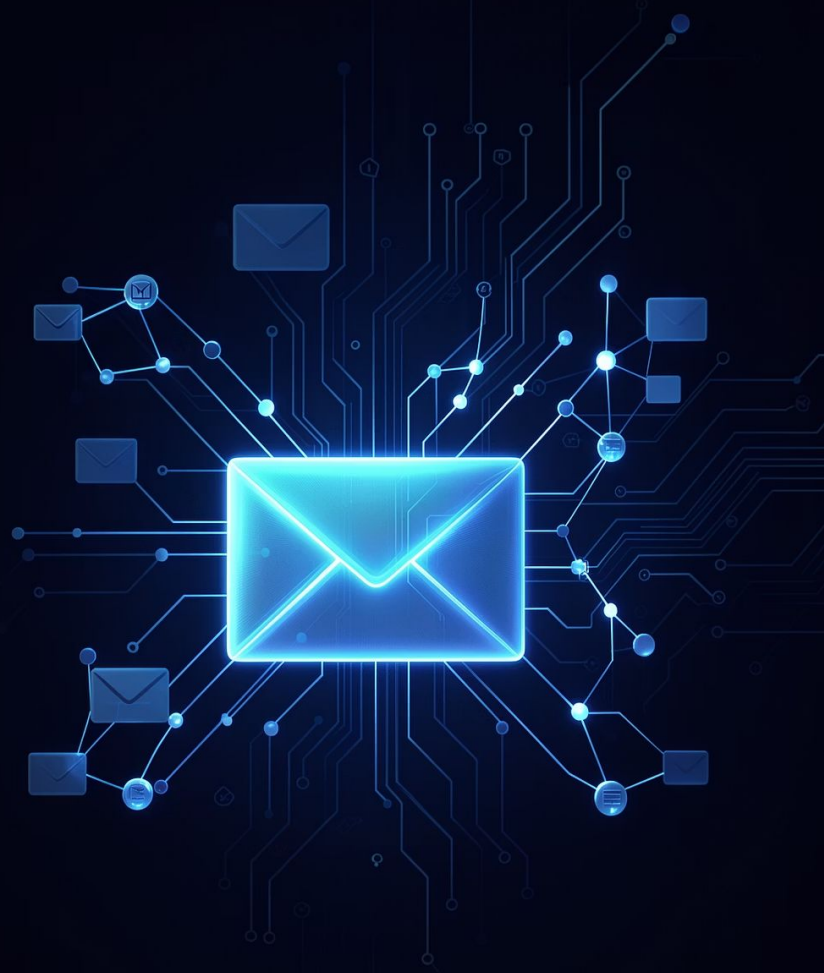
Spam Detection System Using Machine Learning

Using Naive Bayes & Natural Language Processing

Presented by: **Shagun Pandey**

Department: CSE – Artificial Intelligence & Machine Learning

United College of Engineering and Research



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Introduction & Objectives

What is Spam vs Ham?

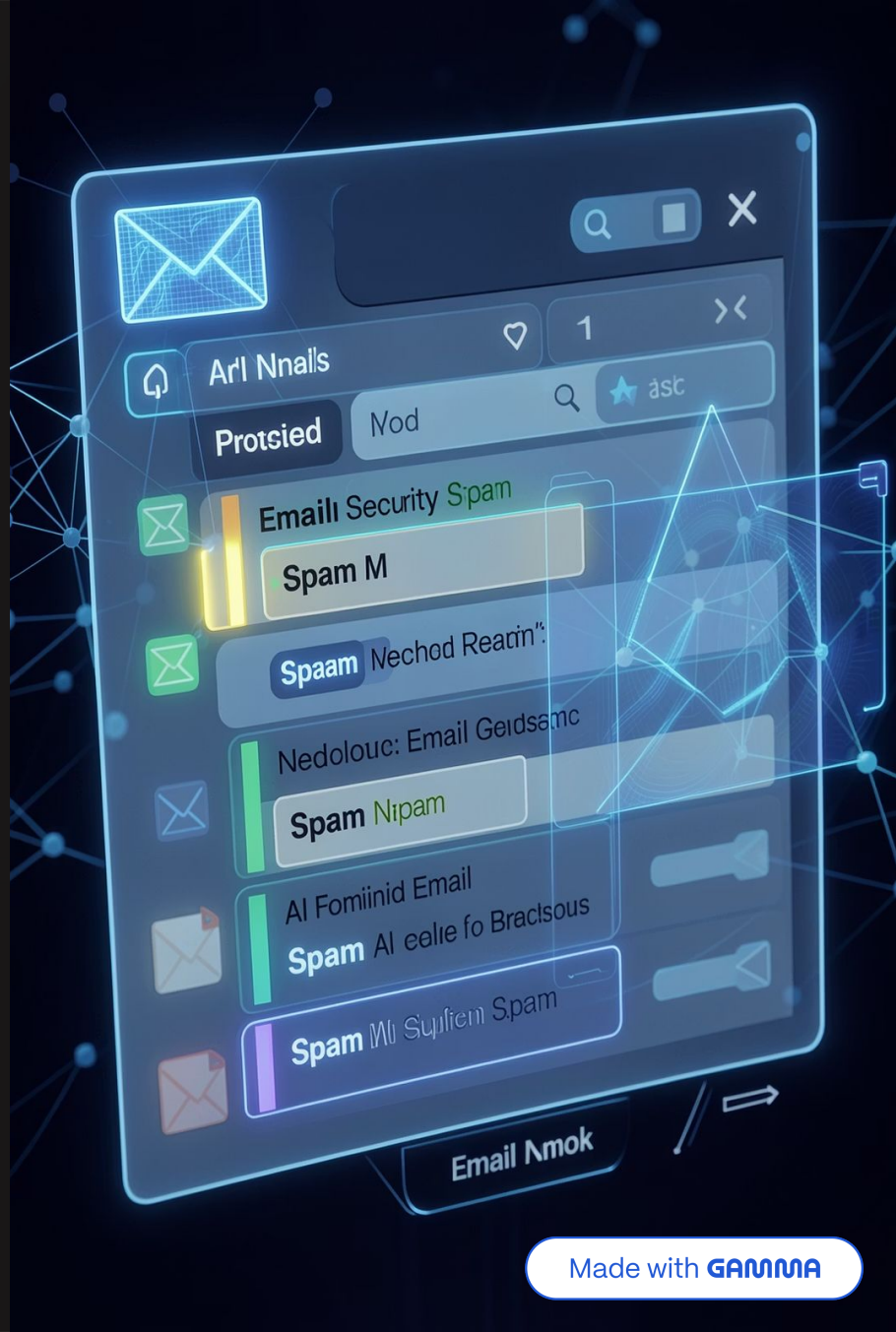
Spam: Unwanted, unsolicited messages often containing ads, scams, or phishing attempts. Ham: Legitimate, relevant messages that users want to receive.

Why Detect Spam?

Protect users from fraud, reduce inbox clutter, improve productivity, and strengthen communication security across platforms.

Project Objectives

- Automatically classify messages as Spam or Ham
- Demonstrate NLP + Machine Learning pipeline
- Improve communication security with a lightweight model



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Dataset & Preprocessing

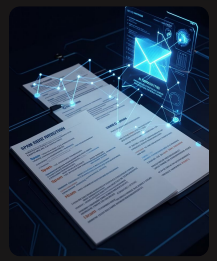
We used the SMS Spam Collection Dataset (5,572 labelled messages). Two classes: **spam** and **ham**. Data is imbalanced and requires careful preprocessing to maximize model performance.

- **Lowercase**
- **Tokenize**
- **Remove Stopwords**
- **Stemming**

The pipeline standardizes text and reduces dimensionality prior to feature extraction (TF-IDF).

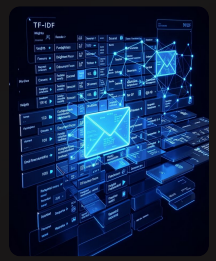
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Methodology



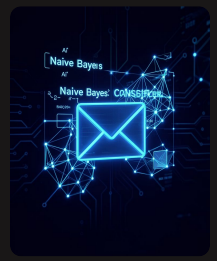
1. Text Preprocessing

Apply tokenization, stopword removal, punctuation cleansing, and stemming to normalize text.



2. Feature Extraction (TF-IDF)

TF-IDF weighs terms by importance across the corpus—reduces influence of common words while highlighting discriminative tokens.



3. Classification: Multinomial Naive Bayes

Chosen for efficiency and strong performance on sparse, text-based features. Works well with TF-IDF and small-to-medium datasets.

Workflow summary: Dataset → Preprocessing → TF-IDF → Multinomial Naive Bayes → Prediction & Evaluation.

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Results & Evaluation

Accuracy

Overall accuracy ≈ **98%** on held-out test set (after preprocessing and hyperparameter tuning).

Classification Metrics

Precision, Recall, and F1 scores were computed per class to measure false positives and false negatives—model shows strong precision for spam detection, with balanced recall.

Confusion Matrix

Low false positive and false negative rates; visualized above for quick inspection.

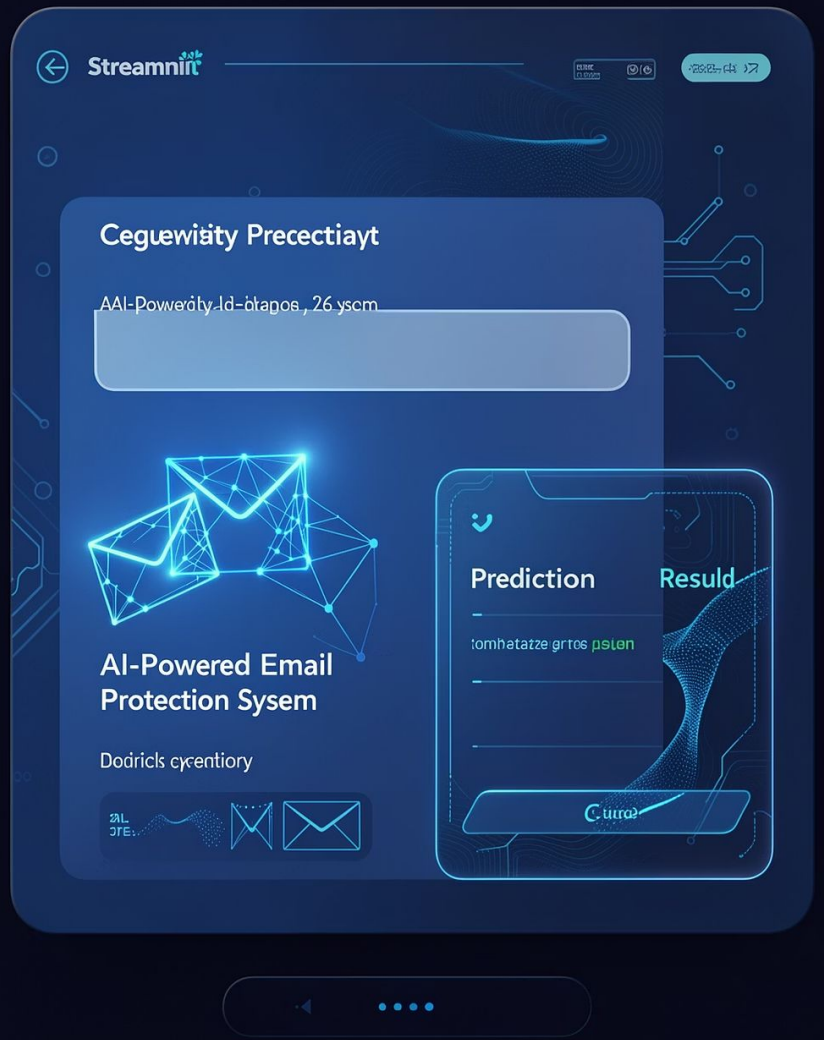
Sample predictions:

"Congratulations! You won ₹10,000" → Spam

"Are we meeting today?" → Ham



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Streamlit Application & Future Scope

A lightweight Streamlit demo was developed to showcase real-time predictions: users paste an SMS/email, the app returns a classification label and confidence score. The app supports model loading, TF-IDF transformation, and displays probability.

Integration points: REST API endpoint for email servers, batch scanning for mailboxes, and a minimal UI for end-user feedback loops.

Future Enhancements

- Transformer models (BERT) and LSTM for contextual understanding
- Multi-language support and extended tokenizers
- Phishing detection, real-time email integration, and adaptive retraining with user feedback

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Conclusion & Thank You

This project demonstrates successful spam classification using NLP preprocessing, TF-IDF features, and Multinomial Naive Bayes. The system achieves high accuracy with efficient inference suitable for real-time applications and further research into deep learning models and multilingual capabilities.

Questions? Contact: **Shagun Pandey** — CSE (AI & ML), United College of Engineering and Research

THANK YOU – Questions?

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